

Diagrammatic Monte Carlo

Assignment #2

Objective: Compute Green's function $G(\tau)$ and observables of different hamiltonians by sampling Feynman diagrams through Monte Carlo simulations.

Computing integrals $\mathcal{Q}(\{y\})$ with Diagrammatic Monte Carlo

With Diagrammatic Monte Carlo (DiagMC) we can compute integrals written in a diagrammatic series of the form

$$\mathcal{Q}(\{y\}) = \sum_{n=0}^{\infty} \sum_{\xi_n} \int \mathcal{D}_n^{\xi_n}(\{y\}, x_1, \dots, x_n) dx_1 \dots dx_n \quad (1)$$

The ingredients of any DiagMC code are:

- **diagrammatic expansion** of $\mathcal{Q}(y)$
- rules to translate diagrams $\mathcal{D}_n^{\xi_n}(\{y\}, x_1, \dots, x_n)$ to numeric expressions
- **updates:** rules to propose a new diagram given the current diagram
 - theoretically: generate and reach any diagram in the expansion $\Rightarrow$ ergodicity
 - Metropolis-Hastings requires detailed balance $\Rightarrow$ each update needs an inverse
- **measurements** during the simulation e.g. :
 - histogram for $\mathcal{Q}(\{y\})$
 - diagram statics

We will see these recipes applied to three cases: `free_propagator`, `potential` and `froehlich` polaron

Hints

The detailed balance equation ensures that the *transition probability* from $x_{curr} \rightarrow x_{new}$ is equal to the one of the inverse transition $x_{new} \rightarrow x_{curr}$. Recalling the expressions seen for the MCMC examples could help in grasping how they change for the Diagrammatic Monte Carlo exercises.

$$f(x_{curr}) p(x_{new}|x_{curr}) = f(x_{new}) p(x_{curr}|x_{new}) \quad (2)$$

The ratio between *acceptance probability* for the current update and the reverse one should satisfy the detailed balance equation:

$$\frac{p_{acc}(x_{new}|x_{curr})}{p_{acc}(x_{curr}|x_{new})} = \frac{f(x_{new})}{f(x_{curr})} \cdot \frac{p(x_{curr}|x_{new})}{p(x_{new}|x_{curr})} \quad (3)$$

Here, the Metropolis choice for the acceptance probability makes Eq.3 easily satisfied. You can check it by considering the reverse update, that means swapping numerator and denominator in the right block of the equation.

$$p_{acc} = \min \left(1, \frac{f(x_{curr}) p(x_{new}|x_{curr})}{f(x_{new}) p(x_{curr}|x_{new})} \right) \quad (4)$$

Free propagator

Free propagator

We compute the imaginary time Green's function $G(\tau)$ of a non-interacting particle (free electron propagator). We check the convergence of the $G(\tau)$ histogram on the number of Monte Carlo samples N for different α and τ .

The Hamiltonian of the exercise is:

$$\hat{H} = \varepsilon \hat{c}^\dagger \hat{c}$$

$G(\tau)$ is the target function. We reconnect to the general formula of the integrals solved with DiagMC Eq.1 for this simple case, where it is just a decaying exponential.

$$Q(\{\tau\}) = e^{\mu\tau} G^0(\tau) = e^{\mu\tau} \langle \emptyset | \hat{c}(\tau) \hat{c}^\dagger(0) | \emptyset \rangle = e^{-(\varepsilon-\mu)\tau} = \mathcal{D}_0(\tau)$$

- $\tau \geq 0$
- ε energy of the particle
- μ artificial chemical potential (simulation parameter)
- τ_{max} max allowed τ value (simulation parameter)

The diagram series is made of only 0th order diagram that looks like:

$$G^0(\tau) = \begin{array}{c} \xrightarrow{\hspace{1.5cm}} \\ 0 \qquad \qquad \tau \end{array} = e^{-(\varepsilon-\mu)\tau}$$

The only update considered in this exercise is the **change- τ** that consist in *stretching* the length of the diagram. The diagrams before and after an accepted **change- τ** updates look like:

$$\mathcal{D}_0(\tau) = \begin{array}{c} \xrightarrow{\hspace{1.5cm}} \\ 0 \qquad \qquad \tau \end{array} \longleftrightarrow \begin{array}{c} \xrightarrow{\hspace{1.5cm}} \\ 0 \qquad \qquad \tau \end{array} = \mathcal{D}_0(\tau')$$

How to do the update in logical steps?

- draw new length τ' uniformly from $\mathcal{U}(0, \tau_{max})$
- compute acceptance probability:

$$P_{acc} = \min \left\{ 1, \frac{\mathcal{D}_0(\tau')}{\mathcal{D}_0(\tau)} \right\} = \min \left\{ 1, e^{-(\varepsilon-\mu)(\tau-\tau')} \right\}$$

As an example, we can monitor the convergence of the sampled $G(\tau)$ for different number of Monte Carlo samples N . Here, the parameters are $\alpha = 1.0$ and $\tau_{max} = 10$

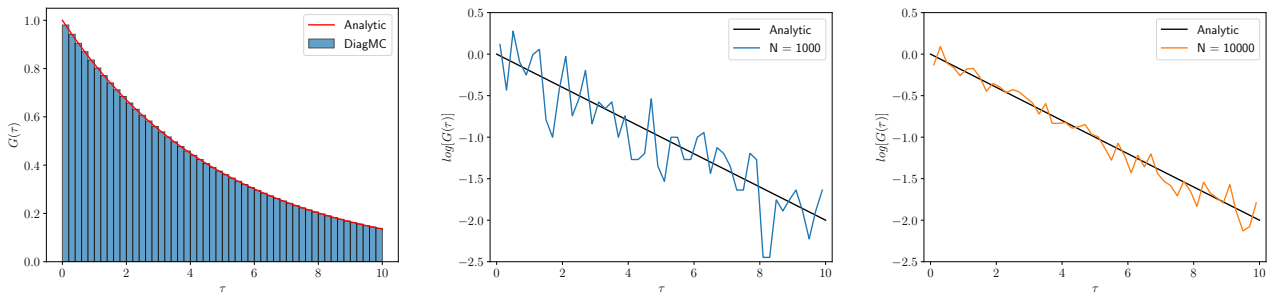


Figure 1: Left: histogram of the sample $G(\tau)$ from DiagMC compared to the exact function. Middle: Log scale plot of $G(\tau)$ sampled with $N=1000$. Right: Log scale plot of $G(\tau)$ sampled with $N=10000$

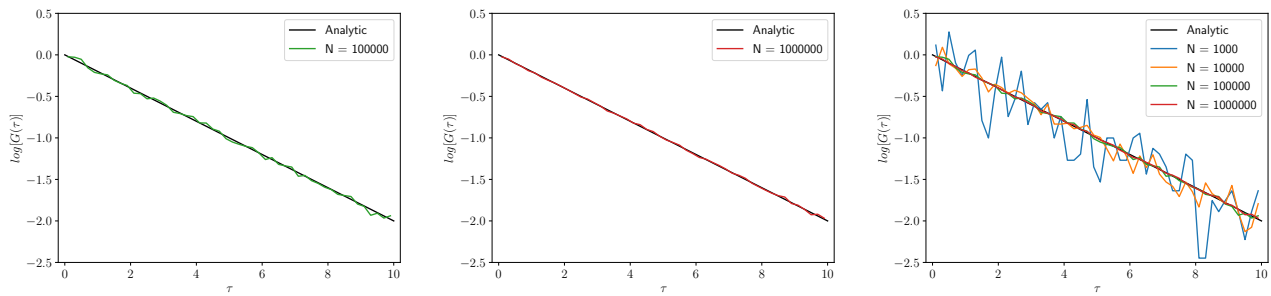


Figure 2: Left: Log scale plot of $G(\tau)$ sampled with $N=100000$ Middle: Log scale plot of $G(\tau)$ sampled with $N=1000000$. Right: Superimposed log scale plot of $G(\tau)$ for different N .

How to?

- Run the code `free_propagator` using the `blade_run` job.
- Check the values written in the input file `input.json`.
- There are several sections such as: `Configuration`, `Simulation`, `Measurements` and `Updates`.

```
{
  "Configuration": {
    "epsilon": 0.0,
    "max_tau": 10.0,
    "mu": -0.3,
    "tau": 1.0
  },
}
```

Configuration: contains the energy of the propagator (`epsilon`), maximum imaginary time (`max_tau`), chemical potential (`mu`) and imaginary time (`tau`) for the first diagram.

```
"Measurements": {
  "greens_function": {
    "grid": {
      "begin": 0.0,
      "end": 10.0,
      "size": 51
    },
    "is_active": true
  }
},
}
```

Measurements: number of bins (`size`) in the time domain between `begin` and `end` for the green's function histogram

```
"Simulation": {
  ...
  "cycles_per_check": 10000,
  "max_steps": 10000,
  "steps_per_cycle": 1,
  ...
},
```

Simulation: technical parameters, such as number of steps, time duration, settings for saving checkpoints etc...e.g. here is a diagmc simulation with 10000 steps.

```
"Updates": {
  "change_tau": {
    "weight": 1.0
  }
},
}
```

Updates: only `change_tau` update for this exercise.

The code `plot_convergence.py` reads the file `greens_function.dat` generated after a successful DiagMC simulation and plot like in Fig.1-2 with $\log[G(\tau)]$ and the exact function.

Electron propagator interacting with potential Δ

Electron propagator interacting with potential Δ

We compute the imaginary time Green's function $G(\tau)$ for an interacting particle with a potential Δ . We check the convergence of the $G(\tau)$ histogram on the number of Monte Carlo samples N for different values of Δ .

When a constant potential is added to a non-interacting particle this results in a constant energy shift. The problem is still simple and the hamiltonian is:

$$\hat{H} = \varepsilon \hat{c}^\dagger \hat{c} + \Delta \hat{c}^\dagger \hat{c} = (\varepsilon + \Delta) \hat{c}^\dagger \hat{c}$$

Recalling Eq.1 the integral can be written easily as decaying exponential with an extra term Δ :

$$\mathcal{Q}(\{\tau\}) = e^{\mu\tau} G^0(\tau) = e^{\mu\tau} \langle \emptyset | \hat{c}(\tau) \hat{c}^\dagger(0) | \emptyset \rangle = e^{-(\varepsilon - \mu + \Delta)\tau} = \mathcal{D}_0(\tau)$$

- $\tau \geq 0$
- ε energy of the particle
- Δ strength of the potential (for now attractive i.e. $\Delta \leq 0$)
- μ artifical chemical potential (simulation parameter)
- τ_{max} max allowed τ value (simulation parameter)

that means a simple DiagMC strategy would be to sample only 0th order diagram as for the example of the **free propagator**

$$G^0(\tau) = \begin{array}{c} \text{---} \rightarrow \text{---} \\ 0 \qquad \qquad \tau \end{array} = e^{-(\varepsilon - \mu + \Delta)\tau}$$

The problem can be approached with 3 different DiagMC strategies. Here, we present the formulation in terms of a full diagrammatic expansion with the potential as an interaction term that makes it suitable for introducing the logic of updates between Feynman diagrams of different order:

$$G(\tau) = \begin{array}{c} \text{---} \rightarrow \text{---} \\ + \\ \text{---} \times \text{---} \times \text{---} \\ + \\ \dots \end{array} + \begin{array}{c} \text{---} \times \text{---} \\ + \\ \text{---} \times \text{---} \times \text{---} \times \text{---} \end{array}$$

$$e^{\mu\tau} G(\tau) = \sum_{n=0}^{\infty} (-1)^n \int_0^\tau d\tau_n \dots \int_0^{\tau_{i+1}} d\tau_i \dots \int_0^{\tau_2} d\tau_1 e^{-(\varepsilon - \mu)\tau_1} \Delta \dots \Delta e^{-(\varepsilon - \mu)(\tau - \tau_n)}$$

we note that there a sign problem only if $\Delta > 0$. Instead, for $\Delta < 0$, $(-1)^n \Delta^n$ are always +.

Updates:

The updates in this exercise consider not only the changing of τ but also the addition and removal or vertex interactions:

- change- τ update:

$$\mathcal{D}_n(\tau) = \begin{array}{c} \text{---} \times \text{---} \times \text{---} \\ 0 \qquad \qquad \tau \end{array} \Rightarrow \begin{array}{c} \text{---} \times \text{---} \times \text{---} \\ 0 \qquad \qquad \tau' \end{array} = \mathcal{D}_n(\tau')$$

Here, the new length τ' is sampled uniformly from $\mathcal{U}(\tau_L, \tau_{max})$ where τ_L is the last (rightmost) vertex in the diagram. The acceptance probability is:

$$P_{acc} = \min \left\{ 1, \frac{\mathcal{D}_n(\tau')}{\mathcal{D}_n(\tau)} \right\} = \min \left\{ 1, e^{-(\varepsilon - \mu)(\tau - \tau')} \right\}$$

- add-vertex update:

$$\mathcal{D}_n(\tau) = \frac{\times}{0} \text{---} \frac{\times}{\tau_L} \text{---} \frac{\times}{\tau_R} \text{---} \tau \longleftrightarrow \frac{\times}{0} \text{---} \frac{\times}{\tau_L} \text{---} \frac{\times}{\tau_{n+1}} \text{---} \frac{\times}{\tau_R} \text{---} \tau = \mathcal{D}_{n+1}(\tau)$$

A time point τ_{n+1} is sampled uniformly from $\mathcal{U}(0, \tau)$, its acceptance probability is:

$$P_{acc} = \min \left\{ 1, \frac{p_{rem}}{p_{add}} \frac{\mathcal{D}_{n+1}(\tau)}{\mathcal{D}_n(\tau) \mathcal{U}(0, \tau)} \right\} = \min \left\{ 1, \frac{p_{rem}}{p_{add}} \Delta(\tau - 0) \right\}$$

where:

$$\frac{\mathcal{D}_{n+1}(\tau)}{\mathcal{D}_n(\tau)} = \Delta \quad \mathcal{U}(0, \tau) = \frac{1}{\tau - 0} \quad (5)$$

- remove-vertex update:

$$\mathcal{D}_{n+1}(\tau) = \frac{\text{X}}{0} \frac{\text{X}}{\tau_L} \frac{\text{X}}{\tau_{n+1}} \frac{\text{X}}{\tau_R} \tau \longleftrightarrow \frac{\text{X}}{0} \frac{\text{X}}{\tau_L} \frac{\text{X}}{\tau_R} \tau = \mathcal{D}_n(\tau)$$

its the inverse update to add-vertex. Here, a vertex is chosen randomly with uniform probability. The acceptance probability is:

$$P_{acc} = \min \left\{ 1, \frac{p_{add}}{p_{rem}} \frac{\mathcal{D}_{n-1}(\tau) \mathcal{U}(\tau_L, \tau_R)}{\mathcal{D}_n(\tau)} \right\} = \min \left\{ 1, \frac{p_{add}}{p_{rem}} \frac{1}{\Delta(\tau_L - \tau_R)} \right\}$$

where τ_L and τ_R are the time points of the vertices to the left and right to the vertex to be removed.

Measurements:

To carry out the histogram $H(\tau_i)$ normalization for $G^0(\tau)$ we apply a slightly different strategy compared to the one seen in MCMC. We count how often 0th order diagrams are generated N_0 and we estimate the normalization for 0th order diagrams.

$$I_0 = \int_0^{\tau_{max}} d\tau \mathcal{D}_0(\tau) = \int_0^{\tau_{max}} d\tau e^{-(\varepsilon-\mu)\tau} = \frac{1 - e^{-(\varepsilon-\mu)\tau_{max}}}{\varepsilon - \mu}$$

$$\implies H(\tau_i) = \frac{\tilde{H}(\tau_i)}{\Delta\tau_i} \frac{I_0}{N_0}$$

The update statistics to monitor concerns *acceptance-rejection* ratio and the number of times each update is sampled.

How to?

- Run the code `attractive_potential` using the `blade_run` job.
- Check the values written in the input file `input.json`.
- Monitor and change the sections: **Configuration**, **Simulation**, **Measurements** and **Updates** when needed.
- The code `plot_fit.py` reads the file `greens_function.dat` generated after a successful DiagMC simulation and make a plot $\log[G(\tau)]$ with the exact function.

Froehlich Polaron

We compute the histogram of $G(\mathbf{k}, \tau)$ and extract E_{GS} and Z_{GS} for different electron wave-vector $\mathbf{k}$ and different e-ph coupling constant α .

The diagrammatic series for the Froehlich polaron Green's functions consist of diagrams whose main elements are: *electron propagators*, *phonon propagators* and *interaction vertices*.

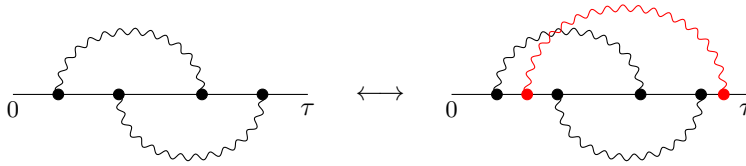
$$G(\mathbf{k}, \tau) = \text{---} + \text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} \text{---} \text{---} + \dots$$

$$e^{\mu\tau} G(\mathbf{k}, \tau) = \sum_{n=0}^{\infty} \sum_{\xi_n} \int_0^{\tau} d\tau_n \dots \int_0^{\tau_{i+1}} d\tau_i \dots \int_0^{\tau_2} d\tau_1 \int \dots \int d\mathbf{q}_1 \dots d\mathbf{q}_{n/2} \mathcal{D}_n^{\xi_n}(\mathbf{k}, \tau, \mu; \tau_1, \dots, \tau_n, \mathbf{q}_1, \dots, \mathbf{q}_{n/2})$$

diagrams $\mathcal{D}_n^{\xi_n}(\mathbf{k}, \tau, \mu; \tau_1, \dots, \tau_n, \mathbf{q}_1, \dots, \mathbf{q}_{n/2})$ are made of electron propagators $e^{-(k_i^2 - \mu)(\tau_j - \tau_i)}$, phonon propagators $e^{-(\tau_j - \tau_i)}$ squared matrix elements $|V_{\mathbf{q}}|^2$ and factor $V/(2\pi)^3$ per wave vector integration.

The updates considered for sampling the Froehlich $G(\mathbf{k}, \tau)$ are:

- change- τ update:
- add/remove vertex updates:



- change vertex- τ update
- rescale-diagram update
- change-phonon-wave-vector-modulus update
- change-phonon-wave-vector-direction update
- change-topology update

The measurements considered during the simulations are:

- histogram $H(\mathbf{k}, \tau_i)$ for $G(\mathbf{k}, \tau)$, normalized using 0th order diagrams
- statistics of accepted and rejected updates
- order of diagrams sampled

The ground state energy $E_{GS}(\mathbf{k})$ and quasiparticle weight $Z_{GS}(\mathbf{k})$ are extracted from Green's function in the limit $\tau \rightarrow \infty$ where it holds:

$$e^{\mu\tau} G(\mathbf{k}, \tau) \xrightarrow[\tau \rightarrow \infty]{\mu < E_{GS}} Z_{GS}(\mathbf{k}) e^{-(E_{GS}(\mathbf{k}) - \mu)\tau}$$

Hints

The choice of μ can be a bit delicate to obtain meaningful results. We report a table and a plot taken from the paper Thomas, 2018. It guide you in choosing reasonable input values for μ ($\mu < E_{GS}$). The DiagMC ground state energies $E_{GS}(0, \alpha)$ computed at different coupling constant α for an electron wave-vector $\mathbf{k} = 0$ are reported in the left table in Fig.3. Optional: to compute $E_{GS}(\mathbf{k}, \alpha)$ at $\mathbf{k} \neq 0$ you can check the energy difference $E_{GS}(\mathbf{k}, \alpha) - E_{GS}(0, \alpha)$ from the plot in Fig.3. There, the coupling values are $\alpha = 1.0, 2.0, 3.0$. Again, you can assign a μ slightly smaller than the $E_{GS}(\mathbf{k}, \alpha)$ estimated from the plot.

For your interest, in the paper Thomas, 2018 you can find extra information regarding the energy dispersion for both the 2D and 3D cases as well as the results for polaron effective masses.

α	E_0 DMC	E_0 Feynman	m_* DMC	m_* Feynman
1	-1.01662(47)	-1.0130308	1.19396(2)	1.1955147
2	-2.06957(84)	-2.0553559	1.46166(7)	1.4718919
3	-3.16829(136)	-3.1333335	1.85047(13)	1.8889540
4	-4.32490(211)	-4.2564809	2.45196(57)	2.5793104
5	-5.55297(296)	-5.4401445	3.47194(180)	3.8856197
6	-6.86647(287)	-6.7108710	5.41952(625)	6.8383564
7	-8.31039(309)	-8.1126875	9.7130(268)	14.394070
8	-9.92206(606)	-9.6953709	20.55(14)	31.569255
9	-11.72535(701)	-11.485786	46.90(78)	62.751527
10	-13.7820(136)	-13.490437	98.8(3.3)	111.81603
11	-16.0660(127)	-15.709808	158.2(4.6)	183.12497
12	-18.5943(240)	-18.143395	270.1(20.0)	281.62189
13	-21.2434(249)	-20.790681		412.78190
14	-24.1151(369)	-23.651278		582.58390
15	-27.2629(359)	-26.724904		797.49838

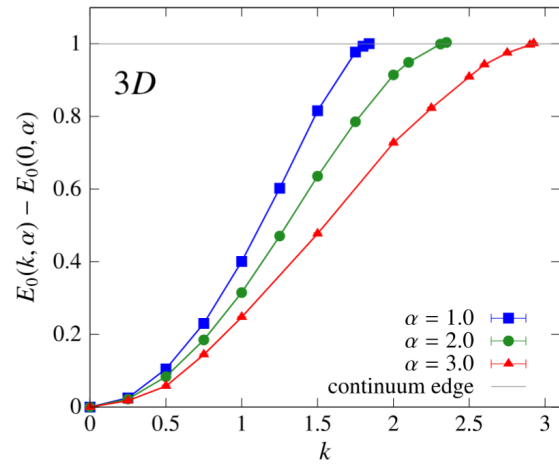


Figure 3: Left: Polaron energies $E_{GS}(0, \alpha)$ for $\mathbf{k} = 0$. Right: Polaron energy dispersion $E_{GS}(\mathbf{k}, \alpha) - E_{GS}(0, \alpha)$ for the values of $\alpha = 1.0, 2.0, 3.0$

Enjoy the lab!

Cesare Franchini, Marco Barducci, Davide Patricelli